POWERING
POTENTIAL

Python Programming Basics

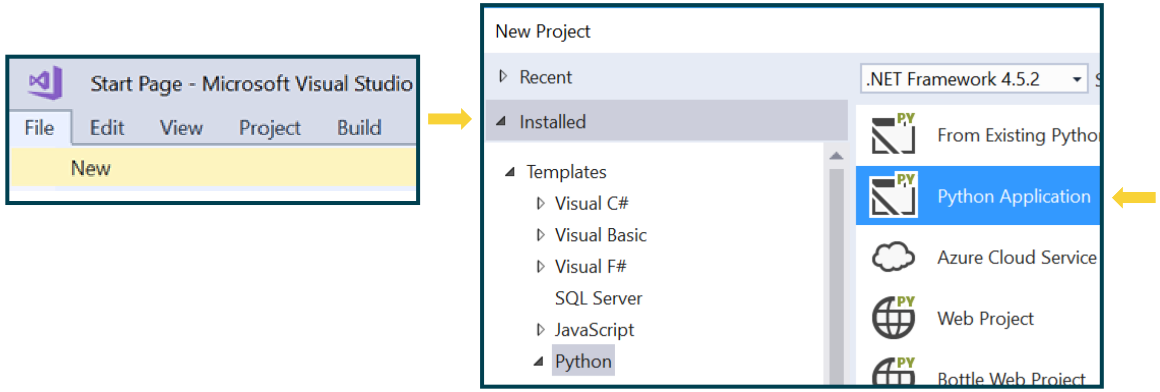
PYTHON FUNDAMENTALS

In this chapter, you'll learn about:

- Python in Visual Studio
- Basic statements
- Numbers, strings and Boolean variables
- Keyboard input
- Screen output
- Casting

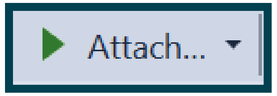
WHY VISUAL STUDIO?

- VS is a superb development environment
  - Excellent debugging and testing facilities

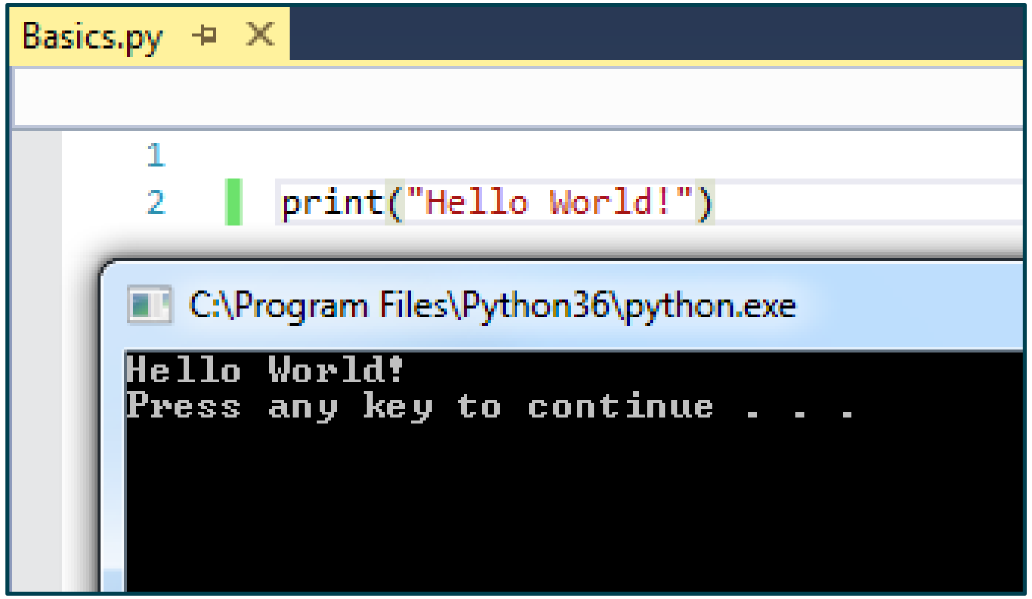


- There are many online compilers
  - https://repl.it/languages/python3

PRINTING TO CONSOLE WINDOW

- Print ('Hello World')
- Click  or F5 key to run

- Press *Ctrl*-F5 to run without debugging



COMMENTS IN CODE

- Help readers understand your code
- Use a # to ignore the following chars
- Are ignored when code runs

```
# process user information

print('Hello World!') # display greetings
```

DATA TYPES IN PYTHON

There Are Three Basic Variable Types:

- Numbers: Integer and Float
  - **1,2,3, 1.23, 0.0005**
- Character or String
  - **'Hello world’ or "Hello world"**
- Boolean **True** or **False** (case-sensitive)

```
age=21
salary = 2000.78
companyName='QA Ltd'
isRegistered = True
hasLicence = False
```

type is determined automatically; value can change

VARIABLE NAMING STANDARDS

- Use letters, not punctuation

```
salay$ = 1500
my-city='London'
my city='London
```

```
my_city='London'
myCity='London'
```

- No 'reserved word'

```
print = 10
```

VARIABLE NAMING STANDARDS

Case Sensitive

- Use lowercase letters for consistency

What will be displayed?

```
age=32
Age=21
print(age)
```

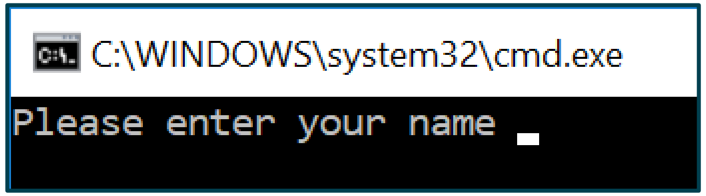
32

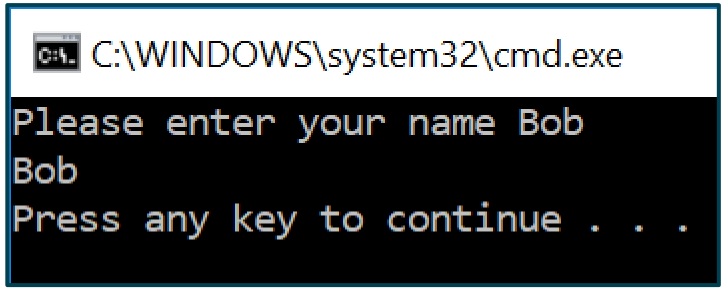
USER INPUT USING KEYBOARD

Input with a Prompt

input(<*prompt*>)

```
name = input('Please enter your name ')
print(name)
```





PUTTING STRINGS TOGETHER

```
username = 'Bob'
print('Hello' , username)
print('Hello' + username)
print('Hello ' + username)
```

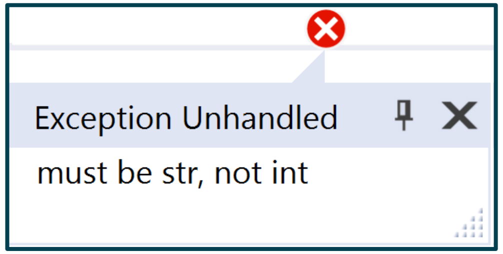
```
Hello Bob
HelloBob
Hello Bob
```

Cannot add numbers and strings

```
age = 21
print('Your age is ' + age)
Message = 'Your age is ' + age
```

**keyboard input is always text…
even if it 'looks' like a number**

```
age = input('Please enter your age ')
age = age + 1
```



CASTING

Keyboard Is Text, So We Use Casting to Convert It to Other Types

```
age = int(input('What is your age? '))
age = age + 1
print('Next year you will be', age ,'years old')
```

```
What is your age?  21
Next year you will be 22 years old
```

```
age = input('What is your age? ')
age = int(age)
age = age + 1
print('Next year you will be', age ,'years old')
```

CASTING A NUMBER TO STRING

Use the str() function

```
## Find the average of a few numbers

total = 1 + 3 + 5 + 7 + 9 + 11
average = total / 6

print("Total is = " + str(total))
print("Average is = " +
str(average))
```

CASTING FLOATS

```
price = int(input('What is the price? '))

totalPrice = price * 1.2
```

```
price = float(input('What is the price? '))

totalPrice = price * 1.2
```

Guided Task

- In this task, you will write code to:
  - Edit code in Visual Studio
  - Compile and run your code
  - Input values into variables
  - Lab duration: 30 minutes

- Please open the instructions in cloud academy

SUMMARY

In this chapter you learned about:

- Python in Visual Studio
- Basic statements
- Numbers, strings and Boolean variables
- Keyboard input
- Screen output
- Casting

FURTHER READING

- https://www.python.org/
- https://www.python.org/dev/peps/pep-0008/#a-foolish-consistency-is-the-hobgoblin-of-little-minds

Thank you!

Hope you enjoyed this learning journey